

ME8930 AI for Composites and Manufacturing

Course Projects

In this course, each student must complete an individual course project. Project topics may be selected from (but are not limited to) the options below:

1. **ML for composite lamina (forward + inverse):** Develop and implement a machine learning model that (i) predicts the anisotropic elastic constants and strengths of a fiber-reinforced polymer composite lamina from the fiber/matrix materials, fiber volume fraction, and fiber orientation, and (ii) performs inverse design to identify lamina designs that achieve user-specified elastic constants and strengths.
2. **ML for composite laminate (forward + inverse):** Develop and implement a machine learning model that (i) predicts the anisotropic elastic constants and strengths of a fiber-reinforced polymer composite laminate from the number of plies, ply thickness, ply-level elastic constants and strengths, and stacking sequence, and (ii) performs inverse design to determine laminate configurations that meet user-specified elastic constants and strengths.
3. **ML for structural performance (forward + inverse):** Build a composite structure using a prescribed set of unidirectional (UD) plies, define a loading case and performance metrics, and develop a machine learning model that (i) predicts structural performance for arbitrary layups and (ii) inversely designs layups to satisfy user-defined performance targets.
4. **ML-enabled process-quality modeling (forward + inverse):** Perform finite element simulations of a composite manufacturing process (e.g., thermoforming, compression molding, injection molding) across a range of process parameters, and develop a machine learning model that (i) predicts manufacturing-induced quality outcomes (e.g., warpage, residual stress) and (ii) inversely identifies process parameters that achieve specified quality requirements.
5. **AI-driven automation for simulation workflows:** Use state-of-the-art AI tools (e.g., Claude Cowork, OpenClaw, ChatGPT Agent, Openwork) to automate elements of a finite element analysis workflow for composite manufacturing, including model setup, simulation execution, post-processing, data generation, and result analysis.
6. **Literature review:** Conduct a comprehensive literature review on AI for composites and manufacturing and produce a review-paper-style report.

Each student is expected to submit a project report and deliver a final presentation. All project data and code must be uploaded to the course Box folder. Reports should be no more than 15 pages, except for the literature review project, which has no page limit.